

$$\rightarrow \bar{X}_i \sim \text{pop}(\mu, \sigma^2)$$

$$\rightarrow \text{Population: } \{x_1, \dots, x_n\} \text{ w/ } \mu, \sigma^2; \text{ Sample: } \{\bar{X}_1, \dots, \bar{X}_n\} \text{ w/ } \bar{\bar{X}}, \sigma^2$$

$$\rightarrow \bar{\bar{X}} \sim \text{normal}(\mu, \sigma^2/n) \text{ (by CLT!)}$$

$$\Rightarrow Z_1 = \frac{\bar{\bar{X}} - \mu}{\sigma/\sqrt{n}} \sim \text{normal}(0, 1)$$

$$\Rightarrow P\left(\frac{\bar{\bar{X}} - \mu}{\sigma/\sqrt{n}} \leq a\right) = \underbrace{\Phi(a)}$$

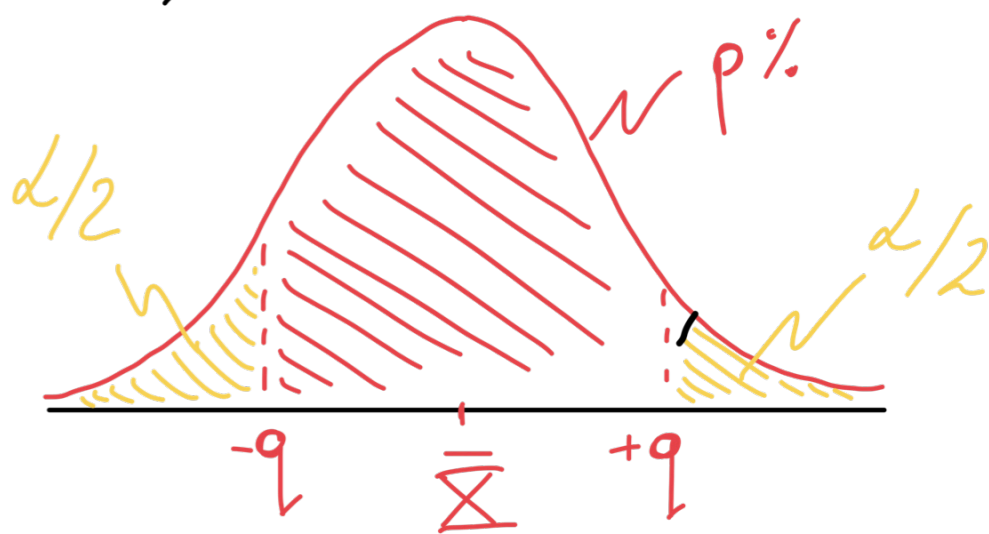
CDF of $N(0, 1)$

$$\rightarrow \text{SAMPLE MEAN: } \bar{\bar{X}} = \frac{1}{n} \sum_{i=1}^n \bar{X}_i$$

CONFIDENCE INTERVAL (for population mean μ)

A $p\%$ confidence interval for μ is a random interval, centered at $\bar{\bar{X}}$, that contains μ w/ probability $p\%$.

$$P(-q \leq \bar{\bar{X}} - \mu \leq q) = p = 1 - \alpha$$



$$P\left(-Z(\alpha/2) \leq \frac{\bar{\bar{X}} - \mu}{\sigma/\sqrt{n}} \leq Z(\alpha/2)\right) = p = 1 - \alpha$$

$$\bar{\bar{X}} \pm \frac{\sigma}{\sqrt{n}} Z(\alpha/2)$$

$\sim p\%$ confidence interval